

Explainable Machine Learning Framework for Early Heart Disease Prediction

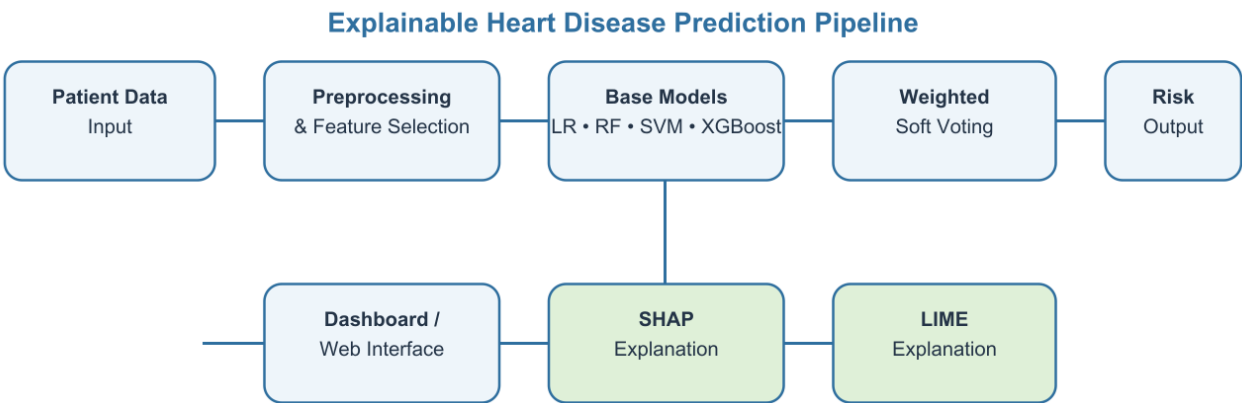
Review 1 Report — Design Phase

Machine Learning / Explainable AI Project

1. System Architecture

The proposed system is designed as an explainable machine-learning decision-support pipeline for early heart disease risk prediction. It combines feature selection, multiple complementary machine-learning models, weighted soft voting, and dual explanation methods to produce a prediction together with human-readable model explanations.

- **Input layer:** Patient clinical attributes such as age, sex, chest pain type, resting blood pressure, cholesterol, ECG-related features, maximum heart rate, exercise-induced angina, oldpeak, slope, CA, and thal are provided to the system.
- **Preprocessing layer:** The dataset is cleaned and prepared for machine-learning analysis before model training and evaluation.
- **Feature Selection layer:** Relevant predictive features are selected before the ensemble stage to reduce unnecessary information and support a more focused prediction pipeline.
- **Base-model layer:** Logistic Regression, Random Forest, SVM, and XGBoost are trained as complementary classifiers.
- **Weighted Soft Voting:** The outputs of the selected base models are combined using weighted soft voting to obtain the final prediction.
- **Explainability layer:** SHAP provides feature-level contribution information, while LIME provides local explanations for individual predictions.
- **Decision-support output:** The system presents the predicted heart-disease risk/class together with explanation information through the user interface.



2. Flow Diagram

Patient Data → Preprocessing → Feature Selection → LR / RF / SVM / XGBoost → Weighted Soft Voting → Prediction → SHAP + LIME Explanations → Dashboard / Web Interface

The flow separates prediction from explanation: the ensemble produces the final classification, while SHAP and LIME provide complementary views of why the model produced that individual result.

3. System Management and Compatibility

- **Runtime:** Python-based machine-learning workflow suitable for development on standard computing hardware.
- **Model libraries:** scikit-learn for classical models and evaluation, XGBoost for gradient-boosted modelling, and SHAP/LIME for explainability.
- **Data handling:** Pandas and NumPy are used for data preparation, feature processing, and numerical operations.
- **Dataset:** The primary dataset is the Cleveland heart-disease dataset, represented in the project as *heart_cleveland_upload.csv*, with 297 records and 14 columns.
- **Interface:** The project can be presented through a Streamlit/Flask-style web interface with HTML/CSS/JavaScript components where required.
- **Modularity:** Preprocessing, feature selection, individual models, ensemble prediction, and explanation modules can be changed independently.

4. Previous Models

Model	Architecture	Main Feature
Logistic Regression	Linear probabilistic classifier	Interpretable baseline for clinical features
Random Forest	Ensemble of decision trees	Captures non-linear feature interactions
Support Vector Machine	Margin-based classifier	Effective for high-dimensional feature spaces
XGBoost	Gradient-boosted decision trees	Captures complex non-linear relationships
Decision Tree	Single decision tree	Simple rule-based prediction structure
Our Proposed Model	Feature Selection + LR + RF + SVM + XGBoost + Weighted Soft Voting	Combines complementary predictions and adds SHAP + LIME explanations

The individual models provide complementary modelling behaviour. The proposed system combines four of them after feature selection through weighted soft voting and adds two explanation mechanisms rather than relying on a single classifier.

5. New Development and Novelty

- **Feature-selected ensemble:** The proposed pipeline combines Logistic Regression, Random Forest, SVM, and XGBoost after feature selection.
- **Weighted soft voting:** Instead of depending on one classifier, the system combines model probability outputs using model weights.
- **Dual explainability:** SHAP and LIME are incorporated so the prediction can be interpreted from both feature-contribution and local-instance perspectives.
- **Patient-oriented presentation:** The project is designed to turn model outputs into understandable risk information rather than displaying only a raw class label.
- **Integrated decision-support workflow:** Prediction, explanation, and dashboard presentation are treated as one practical system.

The project contribution is the integration of feature selection, a weighted ensemble of complementary classifiers, and dual SHAP/LIME explanations into a single early heart-disease prediction workflow.

6. Can Be Executed in Real Life

- **Practical prediction:** A user can enter clinical feature values and receive a model-based heart-disease prediction.
- **Explainable output:** SHAP and LIME can show which input features contributed to an individual prediction.
- **Interactive use:** A browser-based dashboard can present the prediction and supporting explanation without requiring the user to interact directly with Python code.
- **Decision support:** The system can be used as an educational/research decision-support prototype and not as a replacement for professional medical diagnosis.

- **Known limitation:** Performance depends on the dataset, preprocessing, training procedure, and the patient population represented by the data.

7. Offline Deployment

- **Local deployment:** The trained model and preprocessing pipeline can be executed locally without a separate cloud prediction service.
- **Backend:** Python handles preprocessing, model inference, ensemble prediction, and explanation generation.
- **Frontend:** Streamlit or a Flask + HTML/CSS/JavaScript interface can provide the user-facing dashboard.
- **Prediction output:** The interface displays the predicted class/risk result and supporting metrics or confidence information where implemented.
- **Explanation output:** SHAP and LIME visualizations can be displayed alongside the prediction to communicate influential features.

8. Use Cases

- **Early risk screening:** Provide a machine-learning-based risk indication from available clinical attributes.
- **Clinical decision support research:** Demonstrate how ensemble models and explainability can be combined in a healthcare-oriented workflow.
- **Patient-friendly explanation:** Present influential features in a more understandable format.
- **Educational use:** Demonstrate preprocessing, feature selection, ensemble learning, SHAP, and LIME in a real-world classification problem.
- **Research platform:** The modular pipeline can be extended with additional datasets, models, explanation techniques, or evaluation procedures.